Hit Song Prediction Using Audio Features

Machine Learning Mini-Project

1. Problem Statement

Predicting whether a song will become popular is a challenging binary classification problem because popularity can depend on many factors, including musical characteristics and external information. The research paper *"Predicting Hit Songs Using Repeated Chorus"* investigates whether audio characteristics from a repeated chorus can be used to distinguish popular songs from less popular songs.

This project implements an adaptation of that machine-learning framework. The objective is to predict whether a song is a **hit (1)** or **flop (0)** using audio and audio-analysis features, while comparing multiple supervised learning models.

The implementation is an **adaptation rather than an exact reproduction** of the paper. The original study collected songs, extracted repeated 15-second chorus segments, and generated 518 statistical audio features using librosa. Due to the scope of this mini-project, an existing Spotify audio-feature dataset was used instead of reproducing the original audio collection and chorus-extraction pipeline.

2. Dataset

The project uses the **Spotify Audio Features Hit Predictor Dataset (1960–2019)**. The dataset contains songs from six decades: the 1960s, 1970s, 1980s, 1990s, 2000s, and 2010s. Its target variable identifies whether a song appeared on the Billboard Hot-100 weekly list during its respective decade.

After combining the six decade-wise CSV files, the dataset contains **41,106 songs**, with **20,553 hits and 20,553 flops**.

Features: Danceability, Energy, Key, Loudness, Mode, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo, Duration, Time Signature, Chorus Hit, and Sections. Track name, artist, and URI were excluded from model training because they are identifying information rather than musical input features.

3. Methodology and Implementation

The complete pipeline consists of: (1) combining and validating the six datasets; (2) separating the 15 input features from the binary target; (3) creating an 80:20 stratified train-test split; (4) standardizing the input features; (5) applying PCA; (6) training eight classification models; (7) performing 5-fold stratified cross-validation on the training set; (8) evaluating the models on the held-out test set; and (9) generating confusion matrices, ROC curves, and model-comparison plots.

Models: Logistic Regression, Linear Discriminant Analysis with shrinkage, Linear SVM, RBF SVM, Polynomial SVM, Random Forest, Gradient Boosting, and Neural Network.

PCA Analysis

Standardization was followed by PCA with the number of components selected to retain at least 95% of the variance. In this implementation, **12 principal components were retained from the original 15 features**, explaining approximately **96.0% of the total variance**.

The principal-component loadings indicate that different groups of musical characteristics contribute to different directions of variation. For example, PC1 has relatively large contributions from energy, loudness, acousticness, valence, and danceability, while PC2 is strongly associated with duration and number of sections. Other components capture variation associated with key/mode, chorus position, time signature, and instrumentalness. These loadings describe the structure of variation in the feature space and should not be interpreted as direct predictive feature importance.

4. Evaluation and Results

Models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Five-fold cross-validation was used to examine performance stability on the training data, followed by evaluation on an independent held-out test set.

Model	Acc.	Prec.	Recall	F1
Logistic Regression	.7228	.6887	.8131	.7457
LDA	.7173	.6731	.8449	.7493
Linear SVM	.7193	.6759	.8426	.7501
RBF SVM	.7680	.7222	.8712	.7897
Polynomial SVM	.7349	.6744	.9083	.7741
Random Forest	.7615	.7361	.8155	.7737
Gradient Boosting	.7499	.7141	.8340	.7693
Neural Network	.7709	.7391	.8385	.7853

Model	Acc.	Prec.	Recall	F1
Logistic Regression	.7302	.6930	.8268	.7540
LDA	.7235	.6765	.8567	.7560
Linear SVM	.7221	.6746	.8579	.7553
RBF SVM	.7733	.7277	.8735	.7939
Polynomial SVM	.7378	.6770	.9095	.7762
Random Forest	.7693	.7438	.8215	.7807
Gradient Boosting	.7576	.7204	.8419	.7764
Neural Network	.7728	.7498	.8188	.7828

ROC-AUC: Logistic Regression 0.800; LDA 0.798; Linear SVM 0.800; RBF SVM 0.844; Polynomial SVM 0.807; Random Forest 0.841; Gradient Boosting 0.828; Neural Network 0.847.

The cross-validation and test-set results are relatively consistent across the models, suggesting that the observed performance is not substantially different between the training folds and the held-out test set.

5. Conclusion

This project demonstrates a complete machine-learning pipeline for hit-song classification using Spotify audio features. Standardization and PCA reduced the 15-dimensional feature space to 12 principal components while retaining approximately 96% of the variance. Eight classification models were implemented and evaluated using stratified cross-validation and a separate test set.

The results demonstrate that audio and audio-analysis features contain useful information for distinguishing hits from flops, although the task remains challenging and classification performance is far from perfect. The project successfully adapts the central machine-learning methodology of the reference paper while using a publicly available precomputed feature dataset rather than reproducing the paper's original chorus-extraction pipeline.

Future work could move closer to the original study by collecting raw audio, extracting repeated chorus segments, generating the full statistical feature representation, and investigating more advanced audio or deep-learning approaches.

Reference

Predicting Hit Songs Using Repeated Chorus. Research paper provided for the Machine Learning Mini-Project assignment. The original study reports a 554-song dataset with 518 extracted features and evaluates multiple supervised learning models.